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192572092

CSA0802 PYTHON PROGRAMMING

Q1 Find smallest and largest number 10 marks

WAP to find smallest and largest number in an array
input: 12 5 18 3 25 9
output:
Smallest number:3
Largest number:25

Your Code

```
arr = list(map(int, input().split()))

smallest = arr[0]
largest = arr[0]

for num in arr:
    if num < smallest:
        smallest = num
    if num > largest:
        largest = num

print("Smallest number:", smallest)
print("Largest number:", largest)
```

Input

```
12 5 18 3 25 9
```

Output

```
Smallest number: 3
Largest number: 25
```

1. Run Save Submit

Q2 Missing Number 10 marks

WAP to find missing number in an array of shuffled order
input: 4 2 1 6 5
output:3

Your Code

```
arr = list(map(int, input("Enter the array element: ").split()))

n = int(input("Enter the value of n: "))

expected_sum = n * (n+1) // 2
actual_sum = sum(arr)

missing = expected_sum - actual_sum

print("Missing number:", missing)
```

Input

```
3 7 1 2 8 4 5
8
```

Output

```
Enter the array element: Enter the value of n: Missing number: 6
```

2. Run Save Submit

Q3 **Second largest number** 10 marks

WAP to find second largest number in an array
INPUT: 12 5 18 3 25 9
OUTPUT: 18

Your Code

```
arr = list(map(int, input().split()))

arr.sort()

print("Second largest number:", arr[-2])
```

Input

```
12 5 18 3 25 9
```

Output

```
Second largest number: 18
```

Run Save Submit

3.

WAP to print the most repeated number in an array
INPUT: 2 4 6 2 5 3 1 2
OUTPUT:2

Your Code

```
arr = list(map(int, input().split()))

max_count = 0
most_repeated = arr[0]

for i in arr:
    if arr.count(i) > max_count:
        max_count = arr.count(i)
        most_repeated = i

print("Most repeated number:", most_repeated)
```

Input

```
1 3 2 3 4 3 5
```

Output

```
Most repeated number: 3
```

Run Save Submit

4.

Q4 Rotate an array by 'N' positions 10 marks

WAP to rotate an array by 'N' positions

INPUT: 2 5 6 1 3 4

OUTPUT: 6 1 3 4 2 5

Your Code

```python
arr = list(map(int, input("Enter array elements: ").split()))
n = int(input("Enter number of position: "))

n = n % len(arr)

arr = arr[-n:] + arr[:-n]

print("Rotated array:", *arr)
```

Input

```
2 5 6 1 3 4
```

Output

```
Traceback (most recent call
last):
  File
"C:\Windows\TEMP\sandbox_19257209
2RC_6a6267101b96b4.13034542\main.
py", line 2, in <module>
```

Run Save Submit

5.

Q5 Unique elements 10 marks

WAP to print all unique elements in an array

INPUT : 2 4 6 2 5 6 7

OUTPUT : 4 5 7

Your Code

```python
arr = list(map(int, input("Enter array elements: ").split()))

print("Unique elements:")

for i in arr:
    if arr.count(i) == 1:
        print(i, end=" ")
```

Input

```
2 4 6 2 5 6 7
```

Output

```
Enter array elements: Unique
elements:
4 5 7
```

Run Save Submit

6.

Q7 Remove all duplicates in an array

10 marks

WAP to remove all duplicates in an array

INPUT: 2 4 6 2 7 8 3 4

OUTPUT: 2 4 6 7 8 3

Your Code

```python
arr = list(map(int, input().split()))

result = []

for i in arr:
    if i not in result:
        result.append(i)

print(*result)
```

Input

```
2 4 6 2 5 6 7 3
```

Output

```
2 4 6 5 7 3
```

Run Save Submit

7.

WAP to print all 0's followed by non-zero without changing the order

INPUT: 1 0 4 0 2 5 0 7 0 6 0

OUTPUT: 0 0 0 0 0 1 4 2 5 7 6

Your Code

```python
arr = list(map(int, input().split()))

for i in arr:
    if i == 0:
        print(i, end=" ")

for i in arr:
    if i !=0:
        print(i, end=" ")
```

Input

```
10402507060
```

Output

```
10402507060
```

8.

Spaces:2

Your Code

```
s = input("Enter a sentence: ")

vowels = 0
consonants = 0
digits = 0
spaces = 0

for ch in s:
    if ch.lower() in "aeiou":
        vowels +=1
    elif ch.isalpha():
        consonants +=1
    elif ch.isdigit():
        digits+=1
    elif ch == " ":
        spaces +=1
print(" Vowels =", vowels)
print("Consonants =" ,consonants)
print("Digits =", digits)
print("Spaces =", spaces)
```

9.

Input

```
Hello world 123
```

Output

```
Enter a sentence:  Vowels = 3
Consonants = 7
Digits = 3
Spaces = 2
```

Visible Test Cases

Test Case 1

0.00 / 10 marks

Write a python program to check whether the given string is a palindrome
input: malayalam
output: malayalam is a palindrome
input: saveetha
output: saveetha is not a palindrome

Your Code

```
s = input("Enter a string: ")

if s == s[::-1]:
    print("Palindrome")
else:
    print("Not a palindrome")
```

10.

Input

```
malayalam
```

Output

```
Enter a string: Palindrome
```